



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

30-01-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS & TAAS)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	D	A	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C,D	A	B,C,D	A,B,C	B,C	B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	10.00	205.00 to 210.00	4.00	16.00	15.00	2.11 to 2.13
SECTION-II (ii)	Q.	7	8	9			
	A.	25	2	2			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	D	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C,D	A,B,C	B,D	B,C,D	A,D	A,B,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	45.60	30.00	4.00	7.00	5.00
SECTION-II (ii)	Q.	7	8	9			
	A.	2	8	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	C	B	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,C	A,C,D	A,B,C	A,B	A,B,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	720.00	360.00	4.00	0.00	1.00	2.00
SECTION-II (ii)	Q.	7	8	9			
	A.	5	729	7			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

- Ans (A)

$$kA = p$$

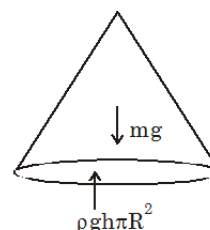
$$\frac{1}{2}m\omega^2 A^2 = q \Rightarrow \frac{1}{2}m\omega^2 \frac{p^2}{k^2} = q$$

$$\frac{1}{2}m\omega^2 \frac{p^2}{m^2\omega^4} = q \Rightarrow \omega = \frac{p}{\sqrt{2mq}}$$
- Ans (D)

$$T \rightarrow \frac{T}{2}$$

$$f \rightarrow 2f$$

3. Ans (A)



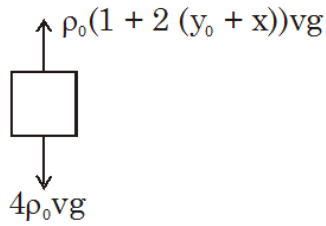
$$\rho gh\pi R^2 = mg + F_{\text{side}}$$

$$\rho gh\pi R^2 = \frac{1}{3}\pi R^2 h \rho g + F_{\text{side}}$$

$$F_{\text{side}} = \frac{2}{3}\pi R^2 h \rho g$$

4. Ans (A)

$$4\rho_0 = \rho_0 (1 + 2y_0) \Rightarrow 3 = 2y_0 \Rightarrow y_0 = \frac{3}{2}$$



$$F_{\text{net}} = -\rho_0 \left(1 + 2 \left(\frac{3}{2} + x \right) \right) vg - 4\rho_0vg$$

$$= -\rho_0 (4 + 2x) vg - 4\rho_0vg$$

$$ma = -\rho_0 2vgx$$

$$4\rho_0v a = -\rho_0 2vgx$$

$$a = -\frac{gx}{2} \Rightarrow \omega = \sqrt{5}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (C,D)

displacement in one time period is 0.

$$v_{\text{avg}} = 0$$

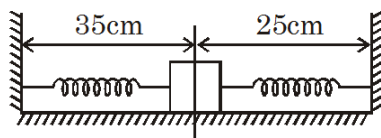
6. Ans (A)

$$k_1x_1 = k_2x_2 \text{ and } x_1 + x_2 = 20$$

$$x_1 = 3x_2$$

$$x_1 = 15 \text{ cm}$$

$$x_2 = 5 \text{ cm}$$



Equilibrium

$$k_{\text{eq}} = 4000$$

$$\omega = \sqrt{\frac{4000}{100}} \times 1000 = 200$$

$$T = \frac{2\pi}{200} = \frac{\pi}{100}$$

7. Ans (B,C,D)

$$v = \sqrt{a - bx^2}$$

$$v = \sqrt{b} \sqrt{\frac{a}{b} - x^2}$$

$$\omega = \sqrt{b} = 2\pi f$$

$$T.E. = \frac{1}{2}kA^2 = \frac{1}{2}mb \left(\frac{a}{b} \right) = \frac{ma}{2}$$

8. Ans (A,B,C)

$$v = \frac{v_0}{H^2} \left(2y^2 - \frac{y^3}{H} \right)$$

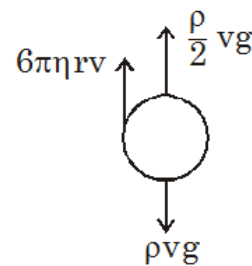
$$\frac{F}{A} = \eta \frac{dv}{dy} = \eta \frac{v_0}{H^2} \left(4y - \frac{3y^2}{H} \right)$$

$$(A) \text{ at } y = \frac{H}{2} ;$$

$$\frac{F}{A} = \eta \frac{v_0}{H^2} \left(2H - \frac{3H}{4} \right) = \eta \frac{5v_0}{4H}$$

$$(B) \text{ at } y = H; \frac{F}{A} = \eta \frac{v_0}{H^2} (4H - 3H) = \frac{\eta v_0}{H}$$

9. Ans (B,C)



$$\frac{\rho}{2}vg = 6\pi\eta rv$$

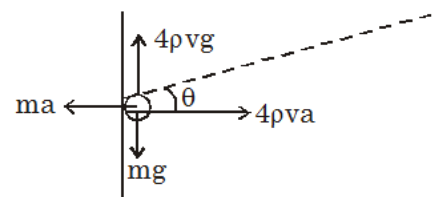
$$\frac{\rho}{2} \frac{4}{3} \pi r^3 g = 6\pi\eta rv$$

$$v = \frac{r^2 \rho g}{9\eta}$$

$$mgH + W = \frac{1}{2} \rho \frac{4}{3} \pi r^3 \left(\frac{r^2 \rho g}{9\eta} \right) - \frac{m}{2} gH$$

$$|W| = \frac{4}{3} \pi r^3 \rho \left[\frac{gH}{2} - \frac{1}{2} \left(\frac{r^2 \rho g}{9\eta} \right)^2 \right]$$

10. Ans (B,C)



$$\tan \theta = \frac{4\rho vg - mg}{4\rho va - ma}$$

$$\frac{1}{2} = \frac{g}{a}$$

$$a = 2g$$

$$\frac{1}{2} a_y t^2 = \frac{L}{2}$$

$$\frac{1}{2} 3gt^2 = \frac{L}{2} \Rightarrow t = \sqrt{\frac{L}{3g}}$$

PART-1 : PHYSICS

SECTION-II (i)

2. Ans (205.00 to 210.00)

$$\frac{dV}{dt} = av = a\sqrt{2gh}$$

4. Ans (16.00)

$$ma = F = \frac{-dU}{dx} = -32(x-1)$$

$$a = -16(x-1) \Rightarrow \omega = 4 \text{ rad/second}$$

At mean position $a = 0 \Rightarrow x = 1$ (mean position)

Applying conservation of mechanical energy

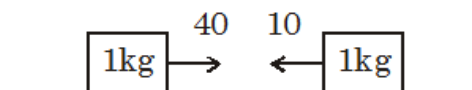
$$0 = -16 + K \cdot \epsilon_{\max} \Rightarrow K \cdot \epsilon_{\max} = 16 \text{ Joule}$$

6. Ans (2.11 to 2.13)

(15 to 16)

$$x = 8 + 4 \sin(10t)$$

$$k = 100 \frac{\text{N}}{\text{m}}$$



$$40 \times 1 - 10 \times 1 = 2v$$

$$30 = 2v$$

$$v = 15$$

$$v = \omega A$$

$$A = \frac{15\sqrt{2}}{10}$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (25)

$$T = 2\pi\sqrt{\frac{\mu}{k}} = 2\pi\sqrt{\frac{6}{5 \times 300}} = 2\sqrt{\frac{6}{5 \times 30}}$$

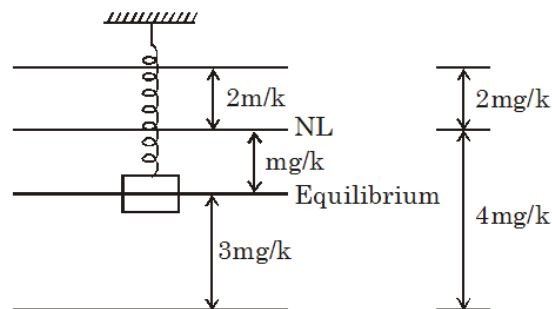
$$T = \frac{2}{5}$$

$$nT = 60$$

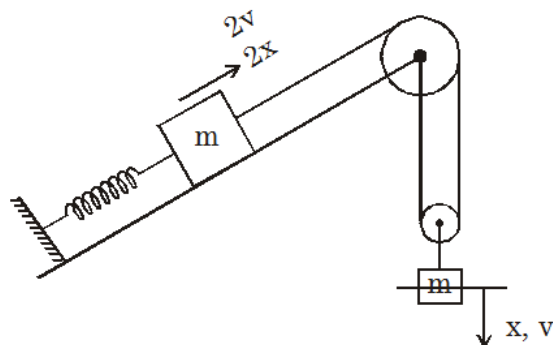
$$n \frac{2}{5} = 60 \Rightarrow n = 150 = 6N$$

$$N = 25$$

8. Ans (2)



9. Ans (2)



$$TE = \frac{1}{2}mv^2 + \frac{1}{2}m(2v)^2 + \frac{1}{2}k(2x)^2$$

$$TE = \frac{1}{2}mv^2 (5) + \frac{1}{2}k(2x)^2 \quad (\text{Using energy method})$$

$$\omega = \sqrt{\frac{4k}{5m}} = 2\sqrt{\frac{k}{5m}}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

7. Ans (B,D)

(I) incorrect M.O.

(II) Hund's rule violation

(III) ground state configuration

(IV) 3 e⁻ in each orbital

PART-2 : CHEMISTRY

SECTION-II (i)

1. Ans (1.50)

$$\rho = \frac{2 \times \frac{450}{6 \times 10^{23}}}{(10 \times 10^{-8} \text{ cm})^3} = 1.5 \frac{\text{gm}}{\text{cc}}$$

PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (2.00)

$$\text{Density} = 8 = \frac{Z_{\text{eff}} \times M}{a^3 \times N_A}$$

$$\Rightarrow M = \frac{8 \times (200 \times 10^{-10})^3 \times 6 \times 10^{23}}{4}$$

$$= 9.6 \text{ g/mol}$$

$$\therefore \text{No. of atoms in 32 gm} = \left(\frac{32}{9.6} \times 6 \times 10^{23} \right)$$

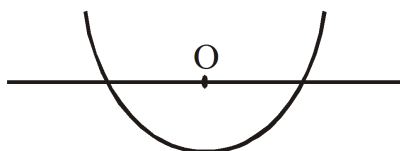
$$= 2 \times 10^{24}$$

$$N = 2$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C)



$$f(x) = e^x + e^{-x} + a$$

$$f'(x) = e^x - e^{-x} = 0$$

$$\Rightarrow x = 0 \text{ is a point of minima}$$

$$\therefore f(0) < 0$$

$$\Rightarrow a + 2 < 0$$

$$\Rightarrow a < -2$$

2. Ans (B)

$$f(8-a) < 0$$

$$\Rightarrow (8-a)(8-a-2a+1) < 0$$

$$\Rightarrow 3 < a < 8$$

$$p = 8 - a \Rightarrow p \in (0, 5)$$

3. Ans (C)

$$x + y + z = 47$$

$\Rightarrow$ The problem is similar to put 47 identical

things in 3 diff. boxes, Let $x' = x - 1$

$$\Rightarrow x' + y + z = 2(23)$$

$$\Rightarrow {}^{23+3-1}C_{3-1} = {}^{25}C_2 = 300$$

4. Ans (D)

Coeff of x^0 in

$$\frac{(x^3 + x^2 + x + 1)^5}{x^5} = \frac{(x^2 + 1)^5 (x + 1)^5}{x^5}$$

$$\therefore \text{Coeff of } x^5 \text{ in } (x^2 + 1)^5 (x + 1)^5$$

$$= \text{Coeff of } x^5 \text{ in } (1 + 5x^2 + 10x^4)$$

$$(1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5)$$

$$= 1 + 50 + 50 = 101$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C)

$$T_9 = {}^{12}C_4 \frac{1}{x^8} \frac{x^8}{2^8} (\log_2 x)^8 = 495$$

$$\Rightarrow \frac{12 \times 1 \times 10 \times 4}{24} \times \frac{1}{2^8} (\log_2 x)^8 = 495$$

$$\Rightarrow \log_2 x = \pm 2$$

$$\Rightarrow x = \frac{1}{4} \text{ or } 4$$

6. Ans (A,C)

$$f(x) = 2x^2 + kx + 50$$

$$|\alpha - \beta| = 8 \Rightarrow \frac{\sqrt{D}}{|a|} = 8 \Rightarrow D = 256$$

$$\therefore \text{Minimum value} = -\frac{D}{4a} = -\frac{256}{8} = -32$$

$$\therefore \text{at } x = 0, f(x) = 50$$

$\Rightarrow$ graph cuts positive y-axis

7. Ans (A,C,D)

$$(1+x)^{100} = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_{100}x^{100}$$

and differentiation $100(1+x)^{99}$

$$= C_1 + {}^2C_2x + {}^3C_3x^2 + \dots + {}^{100}C_{100}x^{99} \dots (i)$$

$$(x+1)^{100} = C_0x^{100} + C_1x^{99} + C_2x^{98} + \dots + C_{100}x^0$$

....(ii)

in (i) $\times$ (ii) coeff of x^{99} is $100 {}^{199}C_{100}$

$$= C_1^2 x^{99} + 2C_2^2 x^{99} + \dots + 100C_{100}^2$$

$$100 {}^{199}C_{100} = 100 {}^{199}C_{99}$$

$$= 100 \times \frac{199!}{99!100!} = \frac{100}{200} \times \frac{200!}{100!100!} \times 100 = 50 {}^{200}C_{100}$$

$$50 {}^{200}C_{100} = 50 \cdot \frac{1.2.3.4 \dots 200}{100! \times 100!}$$

$$= 50 \frac{[1.3.5 \dots 199]}{100!} 2^{100} \frac{(1.2.3 \dots 100)}{100!}$$

$$= \frac{(1.3.5 \dots 199)}{99!} 2^{99}$$

8. Ans (A,B,C)

$$(A) (f+g)(x) = \begin{cases} x^2 + x + 2x + 7 & ; x \geq 4 \\ 2x + 13 & ; 3 \leq x < 4 \\ x^2 - 2x + 10 & ; x < 3 \end{cases}$$

$$\Rightarrow (f+g)(1) = 9$$

$$(B) (f-g)(x) = \begin{cases} -x^2 + 5 & ; x \geq 4 \\ 1 & ; 3 \leq x < 4 \\ x^2 - 4x - 2 & ; x < 3 \end{cases}$$

$$\Rightarrow (f-g)(3.5) = 1$$

$$(C) (f-g)(x) = \begin{cases} (x^2 + x + 2)(x + 7) & ; x \geq 4 \\ (x + 7)(x + 6) & ; 3 \leq x < 4 \\ (x^2 - 3x + 4)(x + 6) & ; x < 3 \end{cases}$$

$$\Rightarrow (f-g)(0) = 24$$

$$(D) \left(\frac{f}{g}\right)(x) = \begin{cases} \frac{x+7}{x^2+x+2} & ; x \geq 4 \\ \frac{x+7}{x+6} & ; 3 \leq x < 4 \\ \frac{x^2-3x+4}{x+6} & ; x < 3 \end{cases}$$

$$\Rightarrow \left(\frac{f}{g}\right)(5) = \frac{12}{32} = \frac{3}{8}$$

9. Ans (A,B)

$$\text{Let } a = 2I - 2, b = 2I, c = 2I + 2$$

$$(A) ac + 4 = b^2$$

$$\Rightarrow \text{LHS} = (2I + 2)(2I - 2) + 4 = 4I^2$$

$$(B) a + c = 2b \Rightarrow (2I - 2) + (2I + 2) = 4I$$

$$(C) a^2 + c^2 = 8b^2$$

$$\Rightarrow \text{LHS} = (2I - 2)^2 + (2I + 2)^2 \neq \text{RHS}$$

$$(D) a + b + c = 6b = \text{not possible.}$$

10. Ans (A,B,C,D)

$$\lfloor 500 \rfloor = k. (2^a 3^b 5^c)$$

$$c = \left\lfloor \frac{500}{5} \right\rfloor + \left\lfloor \frac{500}{5^2} \right\rfloor + \left\lfloor \frac{500}{5^3} \right\rfloor$$

(where $\lfloor . \rfloor$ denotes greatest integer function)

$$= 100 + 20 + 4 \therefore c = 124$$

any set of 2 and 5 produces a zero at the end

$\therefore$ number of zeroes at the end = number of sets

of 2 and 5. Clearly $a > c$

$\therefore$ Number of zeros = 124

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (720.00)

Given word 'HARSHITA'

consonants (H,H), R,S,T vowels $\rightarrow$ (A,A),I

$$\text{Number of ways} = {}^4C_3 \cdot \frac{3!}{2!} \cdot \frac{5!}{2!} = 120 \times 6 = 720$$

2. Ans (360.00)

Given word 'HARSHITA'

consonants (H,H), R,S,T vowels $\rightarrow$ (A,A),I

$$\text{Number of ways} = \frac{6!}{2!} = 360$$

3. Ans (4.00)

$$4^{\log^2 x + 3 \log x} + \frac{64}{4^{\log^2 x + 3 \log x}} = 65$$

$$\text{Let } t = 4^{\log^2 x + 3 \log x}$$

$$\Rightarrow t + \frac{64}{t} = 65$$

$$t = 64 \quad \text{or} \quad t = 1$$

$$4^{\log^2 x + 3 \log x} = 4^3 \quad 4^{\log^2 x + 3 \log x} = 4^0$$

$$\Rightarrow \log^2 x + 3 \log x = 3 \quad \& \quad \log^2 x + 3 \log x = 0$$

$$\Rightarrow A = \text{number of solutions} = 4$$

$$\text{Now, } 3 \sin^2 x + 10 \cos^2 x - 5 = 0$$

$$\Rightarrow 7 \cos^2 x - 2 = 0 \Rightarrow \sec^2 x = \frac{7}{2} = B$$

$$\text{and } \frac{2 \cos 2x \sin 2x - 2 \sin x \cos 3x}{2 \sin 2x}$$

$$\Rightarrow \frac{\sin 4x - \sin 4x + \sin 2x}{2 \sin 2x} = \frac{1}{2} = c$$

4. Ans (0.00)

$$4^{\log^2 x + 3 \log x} + \frac{64}{4^{\log^2 x + 3 \log x}} = 65$$

$$\text{Let } t = 4^{\log^2 x + 3 \log x}$$

$$\Rightarrow t + \frac{64}{t} = 65$$

$$t = 64 \quad \text{or} \quad t = 1$$

$$4^{\log^2 x + 3 \log x} = 4^3 \quad 4^{\log^2 x + 3 \log x} = 4^0$$

$$\Rightarrow \log^2 x + 3 \log x = 3 \quad \& \quad \log^2 x + 3 \log x = 0$$

$$\Rightarrow A = \text{number of solutions} = 4$$

$$\text{Now, } 3 \sin^2 x + 10 \cos^2 x - 5 = 0$$

$$\Rightarrow 7 \cos^2 x - 2 = 0 \Rightarrow \sec^2 x = \frac{7}{2} = B$$

$$\text{and } \frac{2 \cos 2x \sin 2x - 2 \sin x \cos 3x}{2 \sin 2x}$$

$$\Rightarrow \frac{\sin 4x - \sin 4x + \sin 2x}{2 \sin 2x} = \frac{1}{2} = c$$

5. Ans (1.00)

$$(1 - x + x^2)^{30}$$

$$= a_0 + a_1 x + a_2 x^2 + \dots + a_{60} x^{60} \dots (1)$$

$$x \rightarrow -x$$

$$(1 + x + x^2)^{30} = a_0 - a_1 x + a_2 x^2 + \dots + a_{60} x^{60}$$

$$x \rightarrow \frac{1}{x}$$

$$(1 + x + x^2)^{30}$$

$$= a_0 x^{60} - a_1 x^{59} + \dots + a_{60} \dots (2)$$

Multiply (1) and (2)

$$(1 + x^2 + x^4)^{30}$$

$$= a_0 + a_1 x^2 + a_2 x^4 + \dots + a_{60} x^{120} \dots (3)$$

$$(a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots)$$

$$= \text{coeff. of } x^{60} \text{ in } (1 + x^2 + x^4)^{30} = a_{30}$$

$$\left(\frac{a_0^2 - a_1^2 + a_2^2 - a_3^2 + \dots}{a_{30}} \right) = 1$$

6. Ans (2.00)

$$(1 - x + x^2)^{30}$$

$$= a_0 + a_1 x + a_2 x^2 + \dots + a_{60} x^{60} \dots (1)$$

$$x \rightarrow -x$$

$$(1 + x + x^2)^{30} = a_0 - a_1 x + a_2 x^2 + \dots + a_{60} x^{60}$$

$$x \rightarrow \frac{1}{x}$$

$$(1 + x + x^2)^{30}$$

$$= a_0 x^{60} - a_1 x^{59} + \dots + a_{60} \dots (2)$$

Multiply (1) and (2)

$$(1 + x^2 + x^4)^{30}$$

$$= a_0 + a_1 x^2 + a_2 x^4 + \dots + a_{60} x^{120} \dots (3)$$

Put $x = 1$ & -1 (1) add

$$2(a_0 + a_2 + \dots) = 1 + 3^{30}$$

$$\frac{4(a_0 + a_2 + \dots)}{3^{30} + 1} = 2$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (5)

$$2^n = 1024 \Rightarrow n = 10$$

$$\text{number is } 1200 = 2^4 \times 3 \times 25$$

ways are 5

8. Ans (729)

$$(\alpha - 2)^3 \cdot (\beta - 2)^3 \cdot (\gamma - 2)^3$$

$$= ((\alpha - 2)(\beta - 2)(\gamma - 2))^3$$

$\therefore \alpha, \beta, \gamma$ are the roots of the equation

$$x^3 - 5x^2 + 3 = 0$$

$\therefore$ Equation having roots $(\alpha - 2), (\beta - 2), (\gamma - 2)$

$$\text{will be : } (y + 2)^3 - 5(y + 2)^2 + 3 = 0$$

Product of roots of above equation

$$(\alpha - 2)(\beta - 2)(\gamma - 2) = -(8 - 5(4) + 3) = 9$$

$$\therefore ((\alpha - 2)(\beta - 2)(\gamma - 2))^3 = 729$$

9. **Ans (7)**

Let $a = 5^{\alpha_1} \cdot 3^{\beta_1}$

$b = 5^{\alpha_2} \cdot 3^{\beta_2}$

& $c = 5^{\alpha_3} \cdot 3^{\beta_3}$

α_1	α_2	α_3
2	0	3
2	1	3
2	2	3
0	2	3
1	2	3

$\alpha_1, \alpha_2, \alpha_3$ can be selected in 5 ways

where $\beta_1, \beta_2, \beta_3$ can be selected in

13 ways

$\therefore$ Total ways = $5 \times 13 = 65$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS & TAAS)
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

Section-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	C	B	D	D	A	D	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
Section-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	45.00	1.00	0.20	0.25	0.50	1130.00	1.50	15.00	0.03 to 0.04	3.00

PART-2 : CHEMISTRY

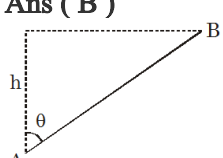
Section-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	D	C	D	B	B	B	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
Section-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	1.00	4.00	13.88 to 13.89	81.00	2.00	5.00	3.00	2.00	4.00	7.00

PART-3 : MATHEMATICS

Section - I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	D	C	A	C	A	B	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
Section-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	7.00	5.00	4.00	8.00	0.00	4.00	309.00	4.00	5.00	7.00

HINT – SHEET
PART-1 : PHYSICS
Section-I

1. **Ans (B)**



$$F_H = \rho \cdot g \cdot \frac{h}{2} \cdot h \cdot b$$

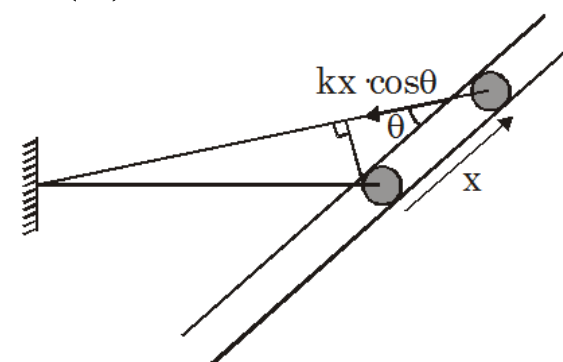
$$= \frac{1}{2} \rho \cdot g h^2 b$$

$$F_v = \frac{1}{2} \cdot h \cdot h \tan \theta \cdot b \cdot \rho \cdot g$$

$$= \frac{1}{2} \rho \cdot g \cdot h^2 b \tan \theta$$

$$F = \sqrt{F_H^2 + F_v^2}$$

$$F = \frac{1}{2} \rho \cdot g h^2 b \cdot \sec \theta$$

2. Ans (A)


$$-(kx \cdot \cos \theta) \cdot \cos \theta = ma$$

$$a = - \left(\frac{k \cos^2 \theta}{m} \right) x$$

$$\omega = \cos \theta \sqrt{\frac{k}{m}}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\cos \theta} \cdot \sqrt{\frac{m}{k}}$$

4. **Ans (B)**

$$F = \rho A$$

$$= \rho \cdot g \cdot 3\ell \cdot A$$

$$= 3\rho \cdot g \ell A$$

5. **Ans (D)**

$$U = U_0 [1 - \cos(bx)]$$

$$F = -\frac{dU}{dx} = -U_0 b \cdot \sin(bx)$$

For small oscillation

$$F = -U_0 b (bx)$$

$$a = -\left(\frac{U_0 b^2}{m}\right)x$$

$$T = \frac{2\pi}{b} \sqrt{\frac{m}{U_0}}$$

6. **Ans (D)**

$$E = \frac{1}{2} m \omega^2 A^2 = \frac{1}{2} m \cdot \frac{k}{m} \cdot A^2 = \frac{1}{2} k A^2$$

$$E' = \frac{1}{2} k A^2$$

$$E' = E$$

7. **Ans (A)**

$$P_0 + \rho g H - \frac{\rho g}{4} \cdot L = P_0$$

$$L = 4H$$

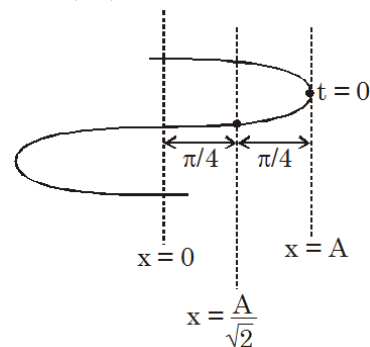
8. **Ans (D)**

$$T = \frac{1}{2} \cdot 2\pi \sqrt{\frac{m}{k}} + \frac{1}{2} \cdot 2\pi \sqrt{\frac{m}{2k}}$$

$$= \pi \sqrt{\frac{m}{k}} + \pi \sqrt{\frac{m}{2k}}$$

$$= \pi \sqrt{\frac{m}{k}} \left(1 + \frac{1}{\sqrt{2}}\right)$$

9. **Ans (B)**



$$\Delta\phi = \omega \cdot \Delta t$$

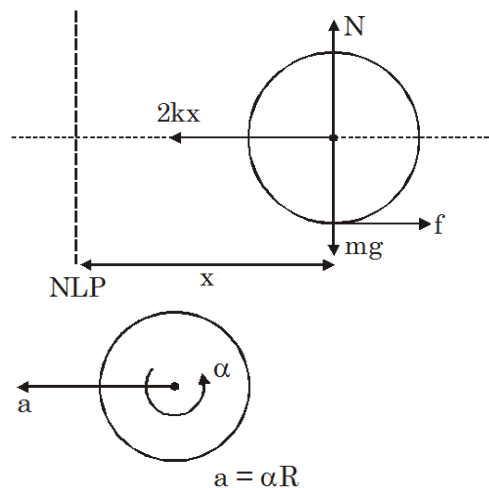
$$= \omega \cdot \frac{\pi}{4\omega}$$

$$= \frac{\pi}{4}$$

$$a = -\omega^2 x$$

$$= -\omega^2 \cdot \frac{A}{\sqrt{2}}$$

10. **Ans (A)**



$$a = \alpha R$$

$$2kx - f = ma \quad \dots (i)$$

$$fR = \frac{2}{5} m R^2 \cdot \alpha$$

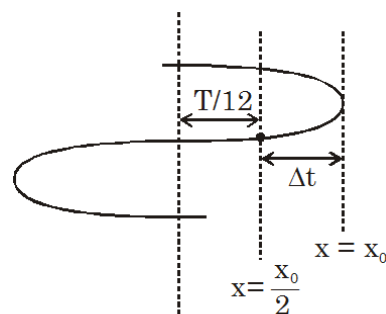
$$f = \frac{2}{5} ma \quad \dots (ii)$$

$$2kx = \frac{7}{5} ma$$

$$a = \frac{10}{7} \frac{kx}{m}$$

$$\omega = \sqrt{\frac{10k}{7m}}$$

$$T = 2\pi \sqrt{\frac{7m}{10k}}$$



$$\Delta t = \frac{T}{4} - \frac{T}{12} = \frac{T}{6}$$

$$= \frac{\pi}{3} \sqrt{\frac{7m}{10k}}$$

11. **Ans (D)**

$$g_{\text{eff}} = \sqrt{g^2 + \left(\frac{v^2}{R}\right)^2} = \sqrt{g^2 + g^2} = g\sqrt{2}$$

$$T = 2\pi \sqrt{\frac{\ell}{g_{\text{eff}}}}$$

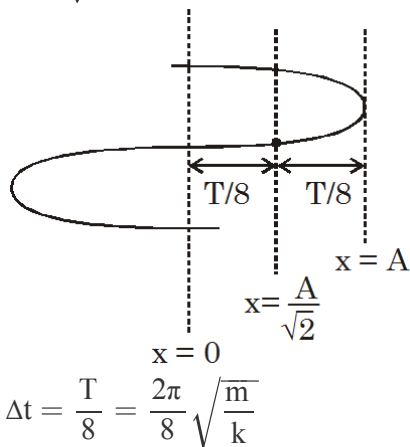
$$= 2\pi \sqrt{\frac{\ell}{g\sqrt{2}}}$$

12. Ans (B)

$$v = \omega \sqrt{A^2 - x^2}$$

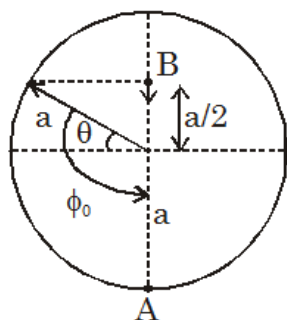
$$\frac{A\omega}{\sqrt{2}} = \omega \sqrt{A^2 - x^2}$$

$$x = \frac{A}{\sqrt{2}}$$



$$\Delta t = \frac{T}{8} = \frac{2\pi}{8} \sqrt{\frac{m}{k}}$$

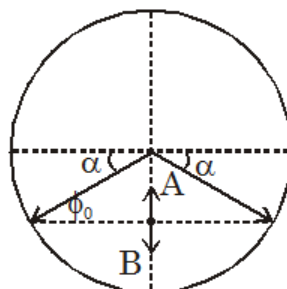
13. Ans (D)



$$\phi_0 = \theta + \pi/2$$

$$\theta = \pi/6$$

$$\phi_0 = \pi/6 + \pi/2 = \frac{4\pi}{6} = \frac{2\pi}{3}$$



$$\theta + \alpha = \omega \Delta t$$

$$\frac{\pi}{2} - \alpha = \omega \Delta t$$

$$\theta + \frac{\pi}{2} = 2\omega \Delta t$$

$$\Delta t = \frac{\theta + \frac{\pi}{2}}{2\omega} = \frac{\frac{2\pi}{3}}{2\omega} = \frac{\pi}{3\omega}$$

14. Ans (C)

$$8 = \frac{1}{2} kA^2$$

$$A = 6 \text{ cm}$$

$$k = \frac{16}{(6 \times 10^{-2})^2} = \frac{16}{36} \times 10^4$$

$$= \frac{40}{9} \times 10^3 = 4.44 \text{ kN/m}$$

mass can't be found $k_{\max}$ at mean.

15. Ans (C)

$$a_{\max} \text{ of } m = \frac{1}{m} (\mu mg \cos \theta - mg \sin \theta) = g \sin \theta$$

$$\Rightarrow (kx - 3mg \sin \theta) = 3m \cdot a_{\max}$$

$$\Rightarrow x = \frac{6mg \sin \theta}{k}$$

$$\text{Amplitude} = x - \frac{3mg \sin \theta}{k} = \frac{3mg \sin \theta}{k}$$

17. Ans (C)

$$I_{CM} = mk^2$$

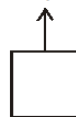
$$\frac{mL^2}{12} = mk^2$$

$$k = \frac{L}{\sqrt{12}}$$

$$\text{Distance} = \frac{L}{2} + \frac{L}{\sqrt{12}}$$

18. Ans (B)

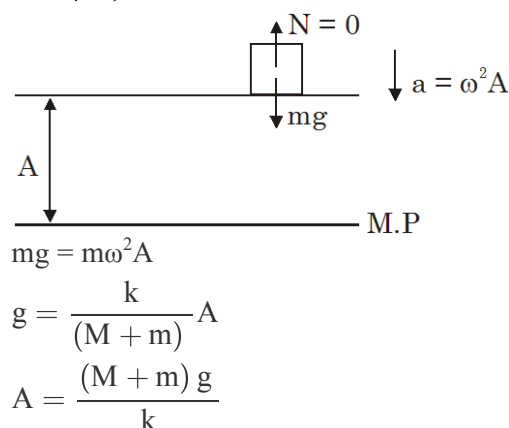
$$F_{\text{net}} = \rho A x g$$



$$F_{\text{net}} = \rho A x g \Rightarrow \frac{\rho}{3} A \ell g \times a = \rho A x g \Rightarrow a = \frac{3x}{\ell}$$

$$\text{time period} = 2\pi \sqrt{\frac{\ell}{3g}}$$

19. Ans (D)



$$g = \frac{k}{(M+m)} A$$

$$A = \frac{(M+m)g}{k}$$

20. **Ans (C)**

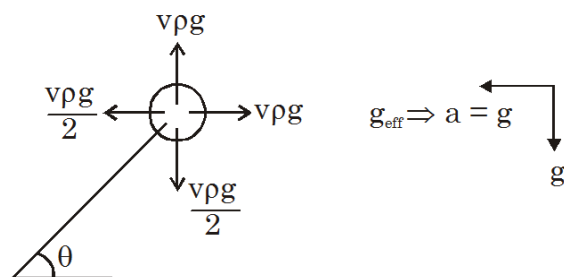
No effect on length inside the liquid, due to motion of container in vertical direction, having constant acceleration.

PART-1 : PHYSICS

Section-II

1. **Ans (45.00)**

$$\tan 45^\circ = \frac{a}{g} \Rightarrow a = g$$



$$\theta = 45^\circ$$

2. **Ans (1.00)**

$$\rho gh = \frac{1}{2} \rho v^2$$

$$v = \sqrt{2gh}$$

$$= \sqrt{2 \times 10 \times \frac{5}{100}}$$

$$= 1 \text{ m/s}$$

3. **Ans (0.20)**

$$t \times v_{\text{max}} = \frac{2\pi}{\omega} \cdot A\omega = 2\pi A$$

$$= 2\pi \times \frac{10}{\pi} \text{ cm}$$

$$= 0.20 \text{ m}$$

4. **Ans (0.25)**

The situation is shown in figure. the moment of inertia of the disc about the wire is

$$I = \frac{mr^2}{2} = \frac{(0.200\text{kg}) (5.0 \times 10^{-2}\text{m})^2}{2}$$

$$= 2.5 \times 10^{-4} \text{ kg-m}^2.$$



The time period is given by

$$T = 2\pi \sqrt{\frac{I}{k}}$$

$$\text{or, } k = \frac{4\pi^2 I}{T^2}$$

$$= \frac{4\pi^2 (2.5 \times 10^{-4} \text{ kg-m}^2)}{(0.20\text{s})^2}$$

$$= 0.25 \frac{\text{kg-m}^2}{\text{s}^2}.$$

5. **Ans (0.50)**

Soap solution film will be formed on both sides of the frame. Each film is in contact with the wire along a distance of 10 cm. The force exerted by the film on the wire

$$= 2 \times (10 \text{ cm}) \times (25 \text{ dyne/cm})$$

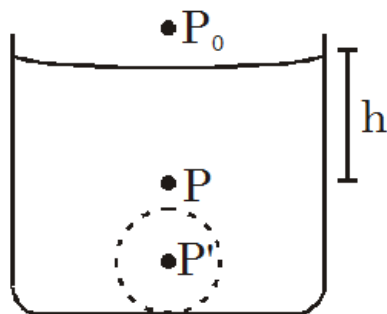
$$= 500 \text{ dyne} = 5 \times 10^{-3} \text{ N}.$$

This force acts vertically upward and should be balanced by the load. Hence the load that should be suspended is

$5 \times 10^{-3} \text{ N}$. The mass of the load should be

$$\frac{5 \times 10^{-3} \text{ N}}{10 \text{ m/s}^2} = 5 \times 10^{-4} \text{ kg} = 0.5 \text{ g}.$$

6. Ans (1130.00)



Let the atmospheric pressure be P_0 . The pressure of the liquid just outside the bubble is (figure)

$$P = P_0 + h\rho g.$$

The pressure inside the bubble is

$$P' = P + \frac{2S}{r} = P_0 + h\rho g + \frac{2S}{r}$$

or,

$$\begin{aligned} P' - P_0 &= (10 \text{ cm}) (1000 \text{ kg/m}^3) (9.8 \text{ m/s}^2) + \frac{2 \times 0.075 \text{ N/m}}{1.0 \times 10^{-3} \text{ m}} \\ &= 980 \text{ N/m}^2 + 150 \text{ N/m}^2 \\ &= 1130 \text{ Pa.} \end{aligned}$$

7. Ans (1.50)

$$P_0 - \frac{2s}{r} + \rho gh + \frac{2s}{r'} = P_0$$

$$\frac{2s}{r'} = \frac{2s}{r} - \rho gh$$

$$\frac{1}{r'} = \frac{1}{r} - \frac{\rho gh}{2s}$$

$$= \frac{1}{0.5 \times 10^{-3}} - \frac{10^3 \times 10 \times 0.02}{2 \times 0.075}$$

$$r' = 1.5 \text{ mm}$$

8. Ans (15.00)

$$\text{For a simple pendulum } T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}}$$

situation 1 : when pendulum is in air

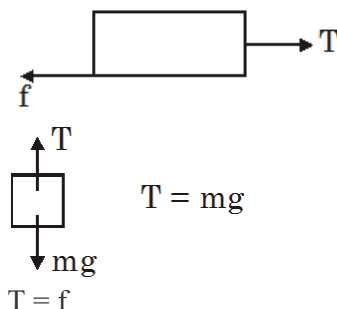
$$\rightarrow g_{\text{eff}} = g$$

situation 2 : when pendulum is in liquid

$$\rightarrow g_{\text{eff}} = g \left(1 - \frac{\rho_{\text{liquid}}}{\rho_{\text{body}}} \right) = g \left(1 - \frac{1}{16} \right) = \frac{15g}{16}$$

$$\text{So, } \frac{T'}{T} = \frac{2\pi \sqrt{\frac{L}{15g/16}}}{2\pi \sqrt{\frac{L}{g}}} \Rightarrow T' = \frac{4T}{\sqrt{15}}$$

9. Ans (0.03 to 0.04)



$$T = mg$$

$$T = f$$

$$mg = \eta \cdot A \frac{\Delta v}{\Delta y}$$

$$\eta = \frac{mg \cdot \Delta y}{A \cdot \Delta v}$$

$$= \frac{0.01 \times 9.8 \times 3 \times 10^{-3}}{0.1 \times 0.085} = 0.03458$$

10. Ans (3.00)

$$\omega = \sqrt{\frac{k}{I}} = \sqrt{\frac{12k}{7mL^2}}$$

$$v = \theta_0 \cdot \omega \cdot \frac{L}{2}$$

$$= \theta_0 \frac{L}{2} \sqrt{\frac{12k}{7mL^2}}$$

$$v = \theta_0 \sqrt{\frac{3k}{7m}}$$

$$\alpha = 3$$

PART-2 : CHEMISTRY

Section-I

2. Ans (B)

$$\text{Volume of 4 spheres} = 4 \times \frac{4}{3} \pi r^3$$

$$\text{Volume of cuboid} = \frac{(4r)^2 \times 2r}{\text{Area} \times \text{height}}$$

Percentage of vacant space =

$$\frac{(4r)^2 \times 2r - 4 \times \frac{4}{3} \pi r^3}{(4r)^2 \times 2r} \times 100 = \left(1 - \frac{\pi}{6} \right) \times 100$$

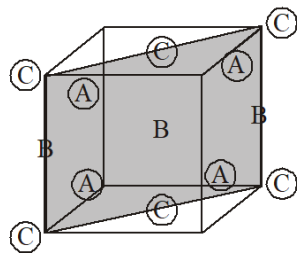
3. Ans (D)

$$\text{Edge length} = 2(r_{\text{Na}^+} + r_{\text{Cl}^-}) = a$$

Shorted distance between Na^+ ions will be along face diagonal

$$= \frac{\sqrt{2}a}{2} = \frac{a}{\sqrt{2}} = \frac{2(95 + 181)}{\sqrt{2}} = 390.3 \text{ pm}$$

4. Ans (C)



C = Corner + Face centres

B = Edge centre (octahedral void) + Body centre (octahedral void)

A = Tetrahedral void



10. Ans (B)

Each iodine is sp^3d^2 hybridised & chlorine is sp^3 hybridised.

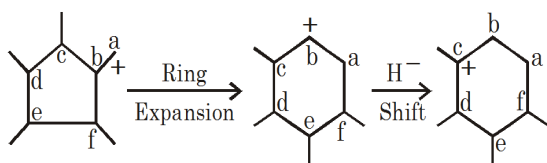
18. Ans (B)

Sulphur is strongest nucleophile due to size. Strong base are good nucleophile (if charge is present on same type of atom)

19. Ans (D)

Leaving group ability $\propto$ its stability
Weak base are good leaving group

20. Ans (D)



PART-2 CHEMISTRY

Section-II

2. Ans (4.00)

$$\frac{r_+}{r_-} = \frac{1.6}{1.864} = 0.858$$

So, it is CsCl type unit cell

$$\text{So } \sqrt{3} a = 2(r_+ + r_-)$$

$$\text{So } a = \frac{2(1.864 + 1.6)}{\sqrt{3}} \text{ \AA} = 2 \times 2 \text{ \AA} = 4 \text{ \AA}$$

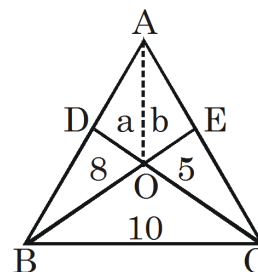
5. Ans (2.00)

SN

PART-3: MATHEMATICS

Section - I

1. Ans (C)



$$\frac{AD}{DB} = \frac{a}{8} = \frac{a+b+5}{18} = \frac{x+5}{18}$$

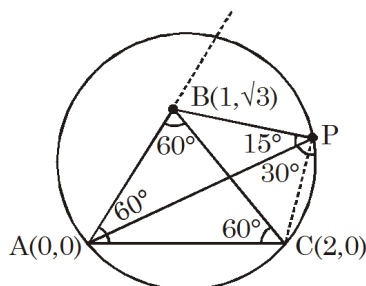
$$\Rightarrow a = \frac{4}{9}(x+5) \quad \dots(i)$$

$$\text{Again, } \frac{AE}{EC} = \frac{b}{5} = \frac{a+b+8}{15} = \frac{x+8}{15}$$

$$\Rightarrow b = \frac{3}{9}(x+8) \quad \dots(ii)$$

$$(i) + (ii) \Rightarrow x = \frac{7x}{9} + \frac{44}{9} \Rightarrow x = 22$$

2. Ans (C)



$$BP = BC = r$$

$$\therefore \angle CBP = 90^\circ$$

$$\therefore BP \perp BC$$

$$\therefore M_{BC} = \frac{\sqrt{3}}{-1}$$

$$\therefore M_{BP} = \frac{1}{\sqrt{3}}$$

3. Ans (D)

$$(|x-2|-2)(|x-2|-3) \leq 0$$

$$2 \leq |x-2| \leq 3$$

$$-3 \leq x-2 \leq -2 \cup 2 \leq x-2 \leq 3$$

$$-1 \leq x \leq 0 \cup 4 \leq x \leq 5$$

$$\text{sum} = -1 + 0 + 4 + 5 = 8$$

4. **Ans (C)**

$$E = \left(x^4 + 4 + \frac{1}{x^4}\right) \left(y^2 + 3 + \frac{1}{y^2}\right) \left((z+1)^2 + 1\right)$$

$$E_{\min} = (6)(5)(1) = 30$$

5. **Ans (A)**

from figure sum of angles of sectors = 2π
 so total distance travelled
 $= 50 + 50 + 20 + 50 + 30 + 2\pi \times 3$

6. **Ans (C)**

$$\left(x + \frac{1}{x}\right) \left(x^2 - \frac{1}{x^2}\right)^9$$

$$\left(x + \frac{1}{x}\right) \sum {}^9C_r (-1)^r x^{18-4r}$$

Coefficient of x^5 is given by $r = 3$

7. **Ans (A)**

$$S = {}^{59}C_{30} + {}^{59}C_{31} + \dots + {}^{59}C_{59} = 2^{58}$$

$$n = 58 = 2 \cdot 29$$

Now $4\lambda + 2 = \text{even number not divisible by 4}$
 and $\lambda \in \mathbb{N}$
 $\Rightarrow 4\lambda + 2 \geq 6$
 $\therefore$ Such type of divisors of 'n' will only be = 58

8. **Ans (B)**

$$(1 - x^2 + x^3) ({}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots) = a_0 + a_1 x + \dots$$

$$a_0 = 1, a_2 = {}^nC_2 - {}^nC_0 = \frac{n^2 - n - 2}{2} = \frac{a_1^2 - a_1 - 2}{2}$$

$$a_1 = n \Rightarrow a_1^2 - a_1 - 2a_2 = 2$$

9. **Ans (C)**

$$T_r = {}^{15}C_{r-1} (x^4)^{15-r+1} \left(-\frac{1}{x^3}\right)^{r-1}$$

$$= {}^{15}C_{r-1} (-1)^{r-1} x^{67-7r}$$

$$67 - 7r = -17 \Rightarrow 7r = 84$$

$$r = 12$$

10. **Ans (D)**

$$\sum_{r=0}^n (r \cdot {}^nC_r)^2 = \sum_{r=1}^n n^2 \left({}^{n-1}C_{r-1}\right)^2 = n^2 \cdot 2^{n-2} C_{n-1}$$

11. **Ans (A)**

$$\frac{n! (r+1)! (n-r-1)!}{(n-r)! r! n!} = \frac{1}{2}$$

$$3r = (n-2)$$

$$\frac{n!}{(n-r-1)! (r+1)!} \times \frac{(r+2)! (n-r-2)!}{n!} = \frac{2}{3}$$

$$5r + 8 = 2n$$

solving $r = 4$ & $n = 14$

12. **Ans (B)**

Best and worst papers occupy cross position in 2 ways

..... $\times$ $\times$

Between these papers there are 3 gaps

Let these gaps are a, b, c

such that

$$a + b + c = 8 \quad \dots(1)$$

$$b \geq 1, a \geq 0, c \geq 0$$

so integral solutions of (i) are 9C_2

and 8 papers are arranged in $8!$ ways

so total permutations are $2 \cdot {}^9C_2 \cdot 8!$

13. **Ans (C)**

$$150 = 2 \cdot 3 \cdot 5^2$$

$$a = 2^{x_1} 3^{y_1} 5^{z_1}$$

$$b = 2^{x_2} 3^{y_2} 5^{z_2}$$

$$c = 2^{x_3} 3^{y_3} 5^{z_3}$$

I) x_1, x_2, x_3 can be assigned in 7 ways.

$$100 \rightarrow 3$$

$$110 \rightarrow 3$$

$$111 \rightarrow 1$$

$$\text{Total } 7$$

II) y_1, y_2, y_3 can be assigned in 7 ways.

III) z_1, z_2, z_3 can be assigned in 19 ways.

$$200 \rightarrow 3$$

$$210 \rightarrow 6$$

$$211 \rightarrow 3$$

$$220 \rightarrow 3$$

$$221 \rightarrow 3$$

$$222 \rightarrow 1$$

$$\text{Total } 19$$

$$\therefore N = 7 \cdot 7 \cdot 19 = 931$$

14. **Ans (A)**

$$p^2 - 3p + 2 = 0 \text{ \& } (p - 1)^2 = 0 \text{ \& } p^2 - 4p + 3 = 0 \\ \Rightarrow p = 1$$

15. **Ans (C)**

$$\left(\frac{\alpha^{\frac{2}{3}}}{\beta^{\frac{1}{3}}} + \frac{\beta^{\frac{2}{3}}}{\alpha^{\frac{1}{3}}} \right) = \frac{\alpha + \beta}{(\alpha\beta)^{1/3}} \\ = \frac{\frac{4}{27}}{\left(\frac{64}{27}\right)^{1/3}} = \frac{4}{\frac{27}{3}} \\ = \frac{3}{27} = \frac{1}{9}$$

16. **Ans (A)**

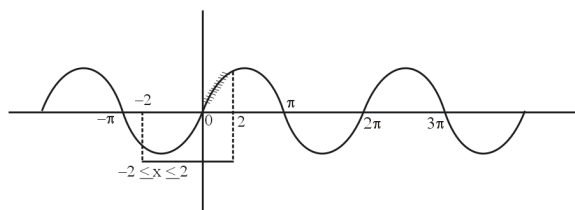
$$y = \frac{11x^2 + 12x + 6}{x^2 + 4x + 2} \\ (y - 11)x^2 + (4y - 12)x + 2y - 6 = 0 \\ D \geq 0 \\ (4y - 12)^2 - 4(y - 11)(2y - 6) \geq 0 \\ 16(y - 3)^2 - 8(y - 11)(y - 3) \geq 0 \\ 8(y - 3)(2y - 6 - y + 11) \geq 0 \\ 8(y - 3)(y + 5) \geq 0 \Rightarrow y \in (-\infty, -5] \cup [3, \infty) \\ y \neq -4, -3, -2, -1, 0, 1, 2$$

17. **Ans (C)**

Domain of $f(x) = \sec^2 x - \tan^2 x$
& $g(x) = 1$ are not same.

18. **Ans (A)**

$$4 - x^2 \geq 0 ; 2 \leq x \leq 2$$



for $f(x)$ to be define $x \in (0, 2]$

19. **Ans (C)**

$$f(x) = \frac{1}{3 \sin 2x - 4 \cos 2x}$$

20. **Ans (A)**

$$\therefore f(x) = \sqrt{\operatorname{sgn} |x|} \\ \therefore \operatorname{sgn} |x| = \begin{cases} 0 & \text{for } |x| = 0 \\ 1 & \text{for } |x| > 0 \end{cases} \\ \Rightarrow f(x) = \begin{cases} 0 & \text{for } x = 0 \\ 1 & \text{for } x \neq 0 \end{cases}$$

PART-3: MATHEMATICS

Section-II

1. **Ans (7.00)**

$$2C_2C_n + 3C_3C_{n-1} + 4C_4C_{n-2} + \dots + nC_nC_2 \\ (1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_{n-1}x^{n-1} + C_nx^n \\ \text{differentiable both sides} \\ n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \dots + nC_nx^{n-1} \\ (1+x)^n = C_0 + C_1x + C_2x^2 + \dots + C_nx^n \\ \text{multiply} \\ n(1+x)^{2n-1} \\ = \dots (2C_0C_n + 3C_3C_{n-1} + \dots + nC_nC_2)x^{n+x} + \dots \\ \text{Coefficient of } x^{n+1} \text{ in } n(1+x)^{2n-1} \\ S(n) \Rightarrow n \cdot {}^{2n-1}C_{n+1} \\ S(8) = 8 \cdot {}^{15}C_9 \\ \frac{8 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10}{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 220 \cdot 26} = 7$$

2. **Ans (5.00)**

$$\text{Let } S = 1 \cdot ({}^{20}C_0)^2 + \dots + 11 \cdot ({}^{20}C_{20})^2 \\ S = 11 \cdot ({}^{20}C_{20})^2 + \dots + 1 \cdot ({}^{20}C_0)^2 \\ 2S = 12 \left[({}^{20}C_0)^2 + ({}^{20}C_2)^2 + \dots + ({}^{20}C_{20})^2 \right] \\ S = 6 \left[\frac{({}^{20}C_0)^2 + ({}^{20}C_1)^2 + \dots + ({}^{20}C_{20})^2}{2} \right] \\ S = 3[{}^{40}C_{20} + {}^{20}C_{10}]$$

3. **Ans (4.00)**

$$\sum_{r=0}^n r \binom{n}{r} = (1) \binom{n}{1} + 2 \binom{n}{2} + \dots + n \binom{n}{n} \\ = n \cdot 2^{(n-1)} \\ S = \frac{1}{2^0} + \frac{2}{2^1} + \frac{3}{2^2} + \frac{4}{2^3} + \dots \dots (1) \\ \frac{S}{2} = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \dots \dots (2) \\ (1) - (2) \\ \frac{S}{2} = 1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \\ \frac{S}{2} = \frac{1}{(1 - \frac{1}{2})} \Rightarrow S = 4$$

4. **Ans (8.00)**

$f(n)$ be the number of ways to get the valuation of rupees 'n'

$$f(n) = f(n-2) + f(n-3) + f(n-4)$$

$$f(2) = 1, f(3) = 1, f(4) = 2$$

Now putting $n = 5, 6, 7, \dots$

we can get all required values

5. **Ans (0.00)**

C-I

$$\underbrace{4}_{\text{inc}} \underbrace{4}_{\text{dic}} \Rightarrow {}^3C_3 \cdot {}^4C_3 \Rightarrow 4$$

C-II

$$\underbrace{9}_{\text{inc}} \dots \Rightarrow {}^8C_3 \cdot {}^9C_3$$

$$\Rightarrow 56 \cdot 84 \Rightarrow 4704$$

$$\Rightarrow 4708$$

6. **Ans (4.00)**

Let two digit number be ab

for $a = b$ condition holds $\Rightarrow 11, 22, \dots, 99$ are 9 numbers.

If $a \neq b$

$$\frac{10a+b}{a} \text{ integer} \Rightarrow \frac{b}{a} \text{ integer} \Rightarrow b = a\lambda$$

$$\frac{10a+b}{b} \text{ integer} \Rightarrow 10 \cdot \frac{a}{b} + 1$$

$$\text{integer} \Rightarrow \frac{10}{\lambda} \text{ integer}$$

$$\Rightarrow \lambda = 2 \text{ or } 5$$

$$\lambda = 2 \Rightarrow b = 2a$$

a	b
1	2
2	4
3	6
4	8

$$\lambda = 5 \Rightarrow b = 5a$$

a	b
1	5

5 more numbers $\Rightarrow 12, 15, 24, 36, 48$.

Total number = 14

7. **Ans (309.00)**

$${}^{10}C_3 + {}^9C_4 + {}^8C_5 + {}^7C_6 = 120 + 126 + 56 + 7 = 309$$

8. **Ans (4.00)**

$$\text{I } 999986 \rightarrow 2 \frac{5!}{4!} = 10$$

$$\text{II } 999887 \rightarrow \frac{5!}{3!} = 20$$

$$\text{III } 998888 \rightarrow \frac{5!}{2!3!} = 10$$

$$\text{Total } 40$$

9. **Ans (5.00)**

$$\sum \frac{(b+c)^2}{a^3-2} = \sum \frac{a^2}{a-1} = \sum a + 1 + \frac{1}{a-1}$$

Now do yourself

10. **Ans (7.00)**

$$\text{We have } (5-\alpha)x^2 - 4x + (5-\alpha) < 0; \forall x \in \mathbb{R}$$

$$\therefore 5-\alpha < 0 \quad \text{Disc} \leq 0$$

$$\alpha > 5 \text{ \& } 16 < 4(\alpha-5)^2$$